CS 215: Task 04: Due: 30-Aug; time as announced on Piazza. No extensions.

- Your answers should be in your handwriting. Try to be neat and clear. If an LLM can read your handwriting, we might be able to.
- Please write the honor code statement only if it is true. If you used any source (person or thing) explicitly state it. This is an individual assignment.

Teaching-team hint.

- When we say, 'derive' or 'show', simply put 2-3 important steps in the derivation in the space provided. This is practice for your midsemester exam, where we may not be able to give extra space. You have to decide what is important. For the homework, please attach extra paper for your records and for you to scan in the upload.

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1. $U \sim \text{Uniform}(0, 1)$ and $a < b$, define

$$U' = a + (b - a)U.$$

   (a) Show that U' is uniformly distributed on (a, b) by working with the cumulative distribution function of U'.

   Teaching-team hint. Do not use any shortcut such as the Leibniz rule discussed in class.

   (b) Derive the moment-generating function of U'.

(c) Use the MGF to compute $E(U')$ and $\mathrm{Var}(U')$.

2. $X \sim \text{Hypergeometric}(M, N, K)$, thus,

$$P(X = x \mid M, N, K) = \frac{\binom{M}{x}\binom{N-M}{K-x}}{\binom{N}{K}}, \qquad x = 0, 1, \ldots, K.$$

In class we discussed the relationship of the expectation and the variance of X with the binomial distribution. Now, show, using Stirling's approximation, viz., $n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$, that if $N \to \infty$, $M \to \infty$, and

$$\frac{M}{N} \to p, \qquad 0 < p < 1,$$

then, for fixed K and x,

$$P(X = x \mid N, M, K) \longrightarrow \binom{K}{x} p^x (1-p)^{K-x}.$$

3. Let X be a random variable with moment-generating function

$$M_X(t) = E(e^{tX}), \qquad -h < t < h.$$

Prove that

$$P(X \geq a) \leq e^{-at} M_X(t), \qquad 0 < t < h.$$

4. $X \sim \text{Binomial}(n, p)$. Our goal is to determine the value of k for which $P(X = k)$, is maximized.

   (a) Show that

$$P(X = k + 1) = \frac{p}{1 - p}\frac{n - k}{k + 1}P(X = k), \qquad k = 0, 1, \ldots, n - 1.$$

   (b) Show that as k goes from 0 to n, the sequence $P(X = k)$ first increases and then decreases. Determine the value of k for which $P(X = k)$ is maximum. Discuss when two values tie.

5. A stick of length 1 metre is broken at two points chosen independently and uniformly at random along its length. Determine the expected lengths of the largest and smallest pieces.

Teaching-team hint. Let the two breaking points, in increasing order, be $U_{(1)}$ and $U_{(2)}$. Express the three piece lengths in terms of these variables. Reason on the constraints in two dimensions.

6. Let $U \sim \text{Uniform}(0,1)$ and define

$$X = U(1-U).$$

Find the cumulative distribution function and probability density function of X.

7. Let $X \sim \text{Binomial}(n,p)$. Define

$$Y = \frac{1}{X+1}.$$

(a) Show that

$$E\left(\frac{1}{X+1}\right) = \frac{1-(1-p)^{n+1}}{(n+1)p}.$$

(b) Deduce a closed-form expression for

$$E\left(\frac{X}{X+1}\right).$$